

Momin Alvi

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EDUCATION

Queen's University

Bachelor of Computing Honours, Software Design Specialization with Professional Internship

Sep 2022 – Apr 2027

Kingston, ON

- **cGPA:** 3.86
- **Relevant Coursework:** Data Structures, Algorithms, OS, Databases, Software Architecture, Linear Algebra

EXPERIENCE

Ontario Teachers' Pension Plan (OTPP)

Software Engineer Intern

May 2025 – Present

Toronto, ON

- Built a **Bash** pipeline that fed **1,000+** high-priority **Dependabot** CVE alerts to a **Claude** agent that resolved them holistically instead of one-by-one, cutting weeks of manual triage down to a few days
- Replaced legacy keyword search with a fully offline **RAG** pipeline (local **Llama** + **ChromaDB**) that answers member pension questions with a cited source URL per answer, keeping data on-machine at \$0 API cost
- Built a **Jenkins CI/CD** pipeline automating **Docker** tagging and registry pushes, reducing deployment steps by **70%**
- Implemented **role-based access control** on a **Spring Boot**/**Stencil** admin app using **Azure** Entitlement Management

Kaniq

AI Systems Developer

Mar 2026 – May 2026

Remote

- Built a **FastMCP** agent server on **GCP Cloud Run** that ranks **100+** banking products to a user's profile with **Gemini 2.5 Pro**, returning the top chequing, savings, and credit-card matches to a downstream AI agent
- Automated catalog refresh with a **Cloud Scheduler** → **Function** → **Cloud Run** pipeline (**OIDC**-secured) that re-scrapes the banks into **MongoDB** with upserts and soft-delete reconciliation
- Scraped **100+** products from **10+** Canadian banks with **Playwright**, normalizing them into a schema with **Gemini**
- Wrote **1,245** pytest cases mocking **Playwright** and **MongoDB** to prevent regressions during pipeline refactoring

Queen's Computing Student's Association (COMPSA)

Backend Developer

Aug 2024 – Dec 2024

Kingston, ON

- Built backend services to automate student club hiring workflows using **Node.js**, **Express.js**, and **Sequelize ORM**
- Designed normalized **Sequelize** models in **MySQL** to improve data integrity and API reliability
- Integrated Calendly and Postmark APIs to automate interview scheduling and email notifications

Dunin-Deshpande Queen's Innovation Centre

Entrepreneur in Residence

May 2024 – Aug 2024

Kingston, ON

- Co-founded Vita Home, a marketplace for senior accessibility equipment to improve home safety and independence
- Built an MVP using **Next.js**, **Node.js**, and **MySQL** to support product discovery and marketplace workflows
- Conducted research with **100+** seniors and professionals, securing **30+** potential customers and validating demand

PROJECTS / OPEN SOURCE

Coursify.ca | *Python, Playwright, Flask, Next.js, TypeScript, Supabase, Gemini* | [Live](#) | [GitHub](#)

Apr 2026 – Present

- Co-built a live course-planning platform for Queen's students with grade data, professor reviews, and schedule planning
- Built a **Playwright/Python** scraper that pulls live class schedules for **320+** courses from Queen's **SOLUS** portal, powering the live schedule planner Queen's students use to build conflict-free timetables
- Built a reusable Supabase upsert pipeline for loading course data into production, validated across **275+** sections
- Contributed to an AI chatbot (**Flask** + **Gemini**) answering course questions from scraped reviews and grade data

Scholar Pups | *FastAPI, Next.js, MongoDB, LangChain, Ollama* | [Devpost](#)

Aug 2024

- Contributed to an AI scholarship platform reducing scholarship search time by **40%**
- Built a **Manifest V3** Chrome extension that injects instant AI feedback on scholarship applications directly into the form
- Served the feedback from a local **Llama 3.1 (Ollama)** + **LangChain** backend via **FastAPI**, with no external API calls
- Scraped government scholarship data with **BeautifulSoup** and **Pandas** into **MongoDB**

TECHNICAL SKILLS

Languages: Java, Python, C, SQL, HTML, CSS, JavaScript, TypeScript, Bash, Haskell

AI/LLM: Gemini 2.5, Vertex AI, LangChain, Llama 3.1, Ollama, RAG, embeddings, ChromaDB

Backend/Cloud: FastAPI, FastMCP, Flask, Node.js, Express.js, Supabase, Upstash Redis, Docker, Jenkins, GCP

Testing/Security: Vitest, Zod, Playwright, Selenium, rate limiting, API validation

Databases/ORMs: MongoDB Atlas, MongoDB, MySQL, Sequelize

Frontend/Tools: React, Next.js, Tailwind CSS, Radix UI, Recharts, Git, Auth0, AEM Cloud, StencilJS, Angular